

INTRODUCTION — Slides 1 & 2 (You)

[Slide 1 — Title]

"Good morning. Dr. Fahad Al-Raddadi, Dr. Hatem Zeini, and our supervisor Dr. Ahmed bin Mahfoudh — thank you for being here with us today.

My name is Abdulrahman Al-dahhas, and I have the honor of leading the team behind this project. On behalf of my six teammates, welcome to the second and final defense of **Eqaab**.

[small pause]

Last semester, we stood in front of you with a vision and a simulation. We had a virtual drone flying in Gazebo, a basic detection prototype, and a system that lived entirely on paper and on a screen. We promised you something more.

Today, we're here to show you that we kept that promise.

This semester, Eqaab is no longer a simulation. It is a real drone — built, trained, and flying. A carbon-fiber airframe carrying a full onboard AI stack, identifying real targets in the air, and sending live GPS-tagged alerts to an operator on the ground.

In one sentence: we took Eqaab from *concept* to *the field*. From an idea we described — to a system you can actually watch fly.

That shift, from simulation to reality, is the story of everything we'll show you in the next twenty-four slides.

[Slide 2 — Agenda]

So let me show you how we'll walk you through it. Our presentation is built around **six chapters**:

We begin with the **Problem Domain** — a quick recap of where we left off, and the real gap we set out to close. Then the **Solution Overview** — our objectives this semester, and how the whole system fits together. After that, the **System Design and Build** — the hardware, the software, and the ground link that make Eqaab fly. Then the **Intelligence Layer** — how Eqaab actually sees and identifies its targets. Then **Validation and Performance** — how we tested it, and what it really delivers in the air. And finally, **Conclusions and Future Work** — what we achieved, and where Eqaab goes next.

To open our first chapter — and to remind us where this journey began — I'll hand it over to my teammate **Omar**. Omar, the floor is yours."

"Thank you, Turki.

Now let me bring everything back to where we started.

Last semester, we made you a promise. Twenty-four weeks ago, Eqaab was an idea on paper — a simulation, a block diagram, and one big question: *can a small student team really build a complete intelligent surveillance system, for real?*

Today, we have our answer. Eqaab is not a slide anymore. It is a real drone that flies on its own, an AI model that sees with **94.3% accuracy**, an IFF system that tells friend from unknown, and a live dashboard that puts all of it in front of an operator in real time.

[look at the committee]

Seven students. Twenty-four weeks. One flying drone — and a system that does the job of infrastructure costing a hundred times more.

But Eqaab was never a one-person project, and I want to be honest about that. Behind every result you saw today, there is a team:

I'd also like to thank our supervisor, **Dr. Ahmed bin Mahfoudh**, for his guidance and trust throughout this journey. And thank you, **Dr. Fahad** and **Dr. Hatem**, for your time and for evaluating our work today.

Before we close, I'd like to share two last things with you.

First — we believe good engineering should be shared. So we made Eqaab **fully open-source**. Every line of code, the trained model, the documentation — all of it is public. If you scan the code on this slide, it will take you straight to our repository. We want the next student team to start where we finished, not from zero.

And second — Eqaab is more than a project to us, it's something we want the world to see. So we built a dedicated **website** for it — a place that tells Eqaab's story, shows the system in action, and introduces the team behind it. The second code on the slide takes you there. We'd be honored if you took a closer look.